

JOUTSA LEIVONMAKI SAVENAHÖ, Finland

WMO: 027710

Lat: 61.879N

Lon: 26.095E

Elev: 479

StdP: 14.44

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.7	-7.1	-17.3	2.2	-12.5	-11.2	3.1	-7.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.3	80.7	65.3	76.7	63.9	73.2	61.8	68.5	76.2	66.3	73.4	64.0	70.2	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.8	96.9	71.3	63.7	89.9	69.4	61.5	83.0	66.7	33.0	76.8	31.2	73.6	29.5	70.6	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
N/A	N/A	N/A	DB	-15.0	84.0	7.7	4.5	-20.5	87.2	-25.0	89.9	-29.3	92.4	-34.9	95.7
			WB	-15.3	70.1	7.6	3.6	-20.8	72.7	-25.3	74.8	-29.5	76.8	-35.1	79.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.6	16.3	20.5	26.8	37.6	50.1	56.6	62.2	58.9	50.4	38.7	32.0	24.3	(6)
	DBStd	17.31	12.48	11.51	8.25	6.49	8.08	6.79	6.40	5.46	5.98	6.41	7.61	10.72	(7)
	HDD50	4845	1043	827	720	375	100	17	3	68	353	541	797		(8)
	HDD65	9334	1508	1247	1185	822	466	264	134	203	437	815	991	1262	(9)
	CDD50	1060	0	0	0	3	102	214	379	279	81	2	0	0	(10)
	CDD65	74	0	0	0	0	3	11	47	13	0	0	0	0	(11)
	CDH74	643	0	0	0	0	57	133	368	85	0	0	0	0	(12)
	CDH80	118	0	0	0	0	5	21	76	16	0	0	0	0	(13)
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(14)
	PrecAvg	25.0	2.0	1.5	1.3	1.3	1.7	2.5	3.2	2.8	2.1	2.4	2.2	2.0	(15)
(16) Precipitation	PrecMax	32.0	3.6	3.3	3.1	2.5	4.5	3.9	7.3	5.4	5.4	5.1	4.6	3.8	(16)
	PrecMin	20.2	0.4	0.1	0.1	0.4	0.4	1.0	0.8	0.7	0.4	0.6	0.2	0.6	(17)
	PrecStd	2.9	0.9	0.8	0.6	0.6	0.9	0.8	1.5	1.3	1.1	1.1	1.0	0.8	(18)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.3	39.9	48.5	65.4	80.0	83.0	86.4	82.2	68.5	55.2	49.2	44.0	(19)
		MCWB	36.6	36.8	39.5	50.5	59.8	64.7	68.3	68.8	60.4	52.4	47.7	42.9	(20)
	2%	DB	34.9	36.5	42.8	58.3	74.6	78.2	81.9	75.6	64.8	52.1	45.7	39.3	(21)
		MCWB	34.2	35.0	36.6	46.6	57.8	62.7	67.2	64.6	58.5	50.2	44.9	38.4	(22)
	5%	DB	33.0	34.6	39.2	53.1	69.9	73.6	78.2	72.0	61.5	50.5	43.2	36.7	(23)
		MCWB	32.6	33.9	35.0	43.5	55.5	60.1	66.2	62.3	56.9	49.0	42.7	36.0	(24)
	10%	DB	31.9	33.0	36.6	48.9	65.4	69.3	74.3	68.4	58.9	48.1	41.0	34.7	(25)
		MCWB	31.6	32.6	33.7	41.6	53.2	58.5	64.5	60.7	56.3	46.8	40.6	34.2	(26)
	0.4%	WB	36.8	37.1	40.9	52.0	62.8	67.5	71.9	70.4	61.7	53.0	48.0	43.1	(27)
		MCDB	37.1	38.8	44.4	62.3	72.8	76.9	80.0	78.4	66.4	54.0	48.5	43.5	(28)
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	34.2	35.4	37.8	48.2	60.1	64.5	69.5	66.7	59.8	51.1	44.9	38.7	(29)
		MCDB	34.5	36.4	41.1	56.0	70.0	73.9	77.3	72.8	63.0	51.6	45.6	39.2	(30)
	5%	WB	32.7	33.9	36.1	44.8	57.6	62.2	67.8	64.3	58.2	49.2	42.7	36.0	(31)
		MCDB	33.0	34.5	38.5	51.5	67.4	70.2	75.3	69.0	60.4	50.1	43.2	36.4	(32)
	10%	WB	31.7	32.7	34.3	42.2	54.7	60.1	65.7	62.3	56.5	46.8	40.7	34.2	(33)
		MCDB	31.9	33.1	36.0	48.0	63.6	67.5	72.5	66.5	58.5	48.0	41.0	34.6	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	9.2	9.9	13.8	16.5	20.2	18.3	17.3	15.7	12.6	9.2	6.0	6.9	(35)
		MCDBR	5.5	7.9	15.9	24.5	27.2	24.3	21.9	20.5	16.3	12.0	8.1	7.1	(36)
		MCWBR	5.4	6.9	11.0	13.7	12.7	10.9	9.6	10.4	10.4	9.4	7.8	7.2	(37)
	5% WB	MCDBR	5.7	7.3	12.8	21.4	23.7	21.5	19.8	16.8	14.0	10.5	8.1	6.9	(38)
		MCWBR	5.8	6.7	9.5	13.0	11.9	10.4	9.8	9.6	9.8	9.3	7.9	7.0	(39)
(40) Clear-Sky Solar Irradiance	taub	0.241	0.288	0.316	0.355	0.353	0.349	0.373	0.358	0.334	0.315	0.262	0.357		(40)
	taud	2.057	2.148	2.225	2.351	2.442	2.479	2.429	2.459	2.516	2.418	2.152	2.622		(41)
	Ebn at Noon	141	211	251	265	276	279	267	262	248	206	132	91		(42)
	Edh at Noon	12	22	29	32	31	31	32	28	22	17	10	7		(43)
(44) All-Sky Solar Radiation	RadAvg	68	249	723	1168	1562	1660	1605	1238	713	293	77	35		(44)
	RadStd	12	58	90	128	174	115	162	146	87	52	14	4		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (51 neighbors)	+0.90	N/A	N/A	N/A	N/A	N/A	-281	N/A	N/A	N/A	(47)

Nomenclature: See separate page